



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FINAL EXAMINATION SEMESTER III SESSION 2021/2022

COURSE CODE : FSPM 0024
COURSE : CALCULUS
PROGRAMME : FOUNDATION UTM
DURATION : 3 HOURS
DATE : APRIL 2022

INSTRUCTIONS TO CANDIDATE:

1. This question paper consists of TWO parts. For both parts, answer ALL Questions.
2. Use a new page for each question. Write clearly using pen.
3. The full marks for each question or section are shown in the bracket at the end of the question.
4. All steps must be shown clearly.
5. Answers may be given in the form of π , e , surd, fractions, or up to four significant figures, where appropriate, unless stated otherwise in the question.
6. If candidate used any methods that have never been discussed in the lectures, they will not be accepted. Any similarities to solutions given in apps will not be accepted.
7. Candidate is not permitted to take screenshots, record the screen, copy and paste questions or answers or otherwise attempt to take any of the content of this exam out of the exam for any purpose.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions according to UTM Academic Regulations.

This examination question consists of (8) printed pages only including this page

PART A (45 MARKS)

There are three questions in this part. Answer all questions.

QUESTION 1 (16 MARKS)

- a) Use the first principle to find the derivative of $f(x) = \frac{1}{\sqrt{4x-1}}$.

(4 marks)

- b) Given a set of parametric equations

$$x = \sin(t^3) \quad \text{and} \quad y = 2 + t^3.$$

Find $\frac{dy}{dx}$ in terms of t . Hence, show that $\frac{d^2y}{dx^2} = x \left(\frac{dy}{dx} \right)^3$.

(6 marks)

- c) By taking natural logarithm on both sides of $y = x^{3 \ln x}$, show that

$$x \frac{dy}{dx} = 6y \ln x$$

by using implicit differentiation. Hence, show that

$$x^2 \frac{d^2y}{dx^2} - x(6 \ln x - 1) \frac{dy}{dx} - 6y = 0.$$

(6 marks)

$$1) a) f(x) = \frac{1}{\sqrt{4x-1}}$$

$$f(x+h) = \frac{1}{\sqrt{4x+4h-1}}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \left(\frac{1}{\sqrt{4x+4h-1}} - \frac{1}{\sqrt{4x-1}} \right) \times \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{4x-1} - \sqrt{4x+4h-1}}{\sqrt{4x+4h-1} \cdot \sqrt{4x-1}} \times \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{4x-1} - \sqrt{4x+4h-1}}{\sqrt{4x+4h-1} \cdot \sqrt{4x-1}} \times \frac{\sqrt{4x-1} + \sqrt{4x+4h-1}}{\sqrt{4x-1} + \sqrt{4x+4h-1}} \times \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(4x-1) - (4x+4h-1)}{4x-1\sqrt{4x+4h-1} + 4x+4h-1\sqrt{4x-1}} \times \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-4h}{4x-1\sqrt{4x+4h-1} + 4x+4h-1\sqrt{4x-1}} \times \frac{1}{h}$$

$$= \frac{-4}{(4x-1)(\sqrt{4x-1}) + (4x-1)\sqrt{4x-1}}$$

$$= \frac{-4}{2(4x-1)(\sqrt{4x-1})} = -2 \left(\frac{1}{(4x-1)^{\frac{3}{2}}} \right)$$

$$(b) x = \sin(t^3)$$

$$u = t^3, \frac{dy}{dt} = 3t^2$$

$$x = \sin u$$

$$\frac{dx}{dt} = \frac{dx}{du} \times \frac{du}{dt}$$

$$= \cos u \times 3t^2$$

$$= 3t^2 \cos t^3$$

$$y = 2 + t^3$$

$$\frac{dy}{dt} = 3t^2$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= 3t^2 \times \frac{1}{3t^2 \cos t^3}$$

$$= \frac{1}{\cos t^3}$$

$$\frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{v u' - u' u}{v^2}$$

$$= \frac{3t^3 \sin t^3}{(\cos t^3)^2}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx}$$

$$= \frac{3t^3 \sin t^3}{(\cos t^3)^2} \left(\frac{1}{3t^2 \cos t^3} \right)$$

$$= \frac{\sin t^3}{(\cos t^3)^3}$$

$$\frac{d^2y}{dx^2} = \frac{\sin t^3}{(\cos t^3)^3} = x \left(\frac{1}{\cos t^3} \right)^3$$

$$= \sin t^3 \cdot \frac{1}{(\cos t^3)^3}$$

$$= \frac{\sin t^3}{(\cos t^3)^3}$$

$$(c) \quad y = x^{3 \ln x}$$

$$\ln(y) = \ln(x^{3 \ln x})$$

$$\ln(y) = 3 \ln x (\ln x)$$

$$\ln(y) = 3 (\ln x)^2$$

$$\frac{d}{dx} \ln(y) = \frac{d}{dx} 3 (\ln x)^2$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx} 3 \ln^2 x \times \frac{d}{dx} \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = 6 \ln x \times \frac{1}{x}$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{6 \ln x}{x}$$

$$x \frac{dy}{dx} = 6 y \ln x$$

$$x \frac{dy}{dx} = 6 y \ln x$$

$$\frac{d}{dx} \left(x \frac{dy}{dx} \right) = \frac{d}{dx} (6 y \ln x)$$

$$u = x \quad v = \frac{dy}{dx} \quad u' = 6 y \quad v' = \ln x$$

$$u' = 1 \quad v' = \frac{d^2 y}{dx^2} \quad u' = 6 \frac{dy}{dx} \quad v' = \frac{1}{x}$$

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} = \frac{6 y}{x} + 6 \ln x \frac{dy}{dx}$$

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} - 6 \ln x \frac{dy}{dx} = \frac{6 y}{x}$$

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - 6 x \ln x \frac{dy}{dx} = 6 y$$

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} (1 - 6 \ln x) - 6 y = 0$$

$$x^2 \frac{d^2 y}{dx^2} - x (6 \ln x - 1) \frac{dy}{dx} - 6 y = 0$$

$$2) (a) \quad p = -6x^3 + 180x^2 - 1350x + 8592$$

$$\frac{dp}{dx} = -18x^2 + 360x - 1350 = 0, \quad x = 15, \quad x = 5$$

$$\frac{d^2 p}{dx^2} = -36x + 360$$

$$\text{when } x = 15, \quad -36x + 360 = -180, \quad p''(x) < 0 \Rightarrow \text{max} \curvearrowright$$

$$\text{when } x = 5, \quad -36x + 360 = 180, \quad p''(x) > 0 \Rightarrow \text{min} \curvearrowleft$$

$$p(15) = 8592$$

$$(b) \quad (i) \quad f(x) = 6x^{\frac{2}{3}}(x-5) = 6x^{\frac{2}{3}}(1) + \frac{4(x-5)}{x^{\frac{1}{3}}} = \frac{6x^{\frac{2}{3}}(x^{\frac{1}{3}})}{x^{\frac{1}{3}}} + \frac{4x-20}{x^{\frac{1}{3}}}$$

$$u = 6x^{\frac{2}{3}} \quad v = (x-5)$$

$$u' = 4x^{-\frac{1}{3}} \quad v' = 1$$

$$= \frac{6x + 4x - 20}{x^{\frac{1}{3}}} = \frac{10x - 20}{x^{\frac{1}{3}}}$$

$$(ii) \quad f'(x) = \frac{10x-20}{x^{\frac{1}{3}}} = 0$$

$$10x - 20 = 0 \quad x^{\frac{1}{3}} = 0$$

$$x = 2 \quad x = 0$$

$$\text{when } x = 2, \quad y = -28.57 \quad x = 0, \quad y = 0$$

QUESTION 2 (12 MARKS)

- a) The total profit of a company for producing and selling x units of merchandise is modelled by

$$P = -6x^3 + 180x^2 - 1350x + 8592,$$

where P is profit in Ringgit Malaysia. Calculate the number of units required to get the maximum profit.

(5 marks)

- b) Given $f(x) = 6x^{\frac{2}{3}}(x - 5)$.

(i) Show that $f'(x) = \frac{10x - 20}{x^{\frac{1}{3}}}$.

(3 marks)

- (ii) Hence, find all the critical points of $f(x)$.

(4 marks)

QUESTION 3 (17 MARKS)

- a) If $\int_0^{\beta} xe^x dx = 4$, use integration by parts to show that $e^{\beta} = \frac{3}{\beta - 1}$.

(4 marks)

- b) Use the substitution of $u^2 = 9x - 2$, where $u > 0$ to evaluate the exact value of

$$\int_{\frac{1}{3}}^{\frac{11}{9}} \frac{81x^2}{\sqrt{9x - 2}} dx.$$

(6 marks)

- c) Using the substitution $u = \ln x$, evaluate

$$\int \frac{18}{x(\ln x - 4)(\ln x + 5)} dx.$$

(7 marks)

$$3) a) \int_0^B x e^x dx = 4 = [x e^x]_0^B - \int_0^B e^x dx = (B e^B - 0) - [e^x]_0^B = B e^B - e^B + 1 = e^B (B - 1) + 1$$

$$u = x \quad \frac{du}{dx} = e^x$$

$$4 = e^B (B - 1) + 1$$

$$\frac{du}{dx} = 1 \quad u' = e^x$$

$$e^B = \frac{3}{B-1}$$

$$(b) \int_{\frac{1}{3}}^{\frac{11}{3}} \frac{81x^2}{\sqrt{9x-2}} dx$$

$$= \int_1^3 \frac{81x^2}{u} \cdot \frac{2u}{9} du$$

$$= \int_1^3 81 \left(\frac{u^4 + 4u^2 + 4}{u^5} \right) \cdot \frac{2}{9} du$$

$$= \frac{2}{9} \left[\frac{u^5}{5} + \frac{4}{3} u^3 + 4u \right]_1^3$$

$$= \frac{2}{9} (96.6 - 5.53)$$

$$= 20.282$$

$$u^2 = 9x-2, \quad x = \frac{u^2+2}{9}, \quad x^2 = \frac{u^4+4u^2+4}{81}$$

$$u = \sqrt{9x-2}$$

$$x = \frac{11}{9}, \quad u = 3$$

$$\frac{du}{dx} = \frac{1}{2}(9x-2)^{-\frac{1}{2}}(9)$$

$$x = \frac{1}{3}, \quad u = 1$$

$$\frac{du}{dx} = \frac{9}{2\sqrt{9x-2}}$$

$$dx = \frac{2\sqrt{9x-2}}{9} du$$

$$= \frac{2u}{9} du$$

$$(c) \int \frac{18}{x(\ln x - 4)(\ln x + 5)} dx, \quad u = \ln x, \quad \frac{du}{dx} = \frac{1}{x}, \quad dx = x du$$

$$= \int \frac{18}{x(u-4)(u+5)} \cdot x du$$

$$= \int \frac{18}{(u-4)(u+5)} du$$

$$= 2 \ln(u-4) - 2 \ln(u+5) + C$$

$$= 2 \ln(\ln x - 4) - 2 \ln(\ln x + 5) + C$$

$$\frac{18}{(u-4)(u+5)} = \frac{A}{u-4} + \frac{B}{u+5} = \frac{2}{u-4} - \frac{2}{u+5}$$

$$A(u+5) + B(u-4) = 18$$

$$A+B=0, \quad 5A-4B=18$$

$$A = -B \quad -9B = 18$$

$$A = 2 \quad B = -2$$